

Intraoperative Methadone Reduces Postoperative Opioid Requirements in Nuss Procedure for Pectus Excavatum

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Abstract

Introduction: Surgical correction of pectus excavatum by Nuss procedure, commonly referred to as minimally invasive repair of pectus excavatum (MIRPE), often results in significant postoperative pain. This study investigated whether adding intraoperative methadone would reduce the postoperative opioid requirement during admission for patients undergoing MIRPE.

Methods: A retrospective cohort chart review was conducted for 40 MIRPE patients between 2018 and 2020. Patients were stratified into 2 groups: those who received multimodal anesthesia (MM, n = 20) and those who received multimodal anesthesia with the addition of intraoperative methadone (MM + M, n = 20). Data collected included total opioid consumption during hospital stay (morphine milligram equivalents [MMEs]), hospital length of stay (LOS), pain scores, time to ambulation, and time to tolerating solid food.

Results: Addition of intraoperative methadone for patients undergoing MIRPE significantly reduced postoperative opioid requirements (MME/kg) during admission ($P = .007$). On average, patients in the MM group received $1.61 \pm .55$ MME/kg while patients in the MM + M group received $1.16 \pm .44$ MME/kg. Hospital opioid (non-methadone) total was also significantly reduced between the MM ($1.87 \pm .54$) and MM + M group ($1.37 \pm .46$), $P = .003$. There was no significant difference in hospital opioid total MME/kg administered between the groups. There were no significant differences observed in hospital LOS, pain scores, time to ambulation, or time to toleration of solid food.

Discussion: Incorporating intraoperative methadone for patients undergoing MIRPE reduced postoperative opioid requirements and hospital opioid (non-methadone) totals without a significant change in pain scores. Patients undergoing the Nuss procedure may benefit from the administration of intraoperative methadone.

Key Takeaways

- The use of intraoperative methadone during MIRPE procedure for repair of pectus excavatum reduces postoperative opioid requirements.
- Patients undergoing MIRPE for repair of pectus excavatum may benefit from intraoperative methadone administration.

Introduction

Ninety percent of congenital chest wall abnormalities are pectus excavatum (PE).¹ The incidence of PE is 3–4% with a male-to-female ratio of 4:1.^{1,2} While the majority of PE is diagnosed before children turn 1 year old, it often becomes symptomatic after the age of 12 years.³ Symptoms of more severe deformities, as measured by

the Haller index, are exercise intolerance (82%), chest pain (68%), poor endurance (67%), and shortness of breath (42%).^{3,4} The minimally invasive procedure for

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repair of PE is known as the Nuss procedure. Dr Donald Nuss published his 10-year experience with the minimally invasive repair in 1998 and it is the most common repair currently performed.⁵ The Nuss procedure is often referred to as the minimally invasive repair of pectus excavatum (MIRPE).

Postoperative pain control is one of the most common causes of prolonged hospitalization after MIRPE, and this pain may persist for up to weeks after surgery.^{6,7} Poor pain management postoperatively may lead to increased anxiety and stress for the patient, increased family stress, and decreased participation in physical and occupational therapies postoperatively, which may subsequently lead to an increased length of stay (LOS) and cost.⁷⁻⁹ Additionally, more morbid complications may develop due to poor pain control such as pneumonia, weight loss, and depression.

Current literature supports the use of multimodal anesthesia in MIRPE which alleviates pain through the blockade of multiple nociceptive and inflammatory pain receptor pathways; multimodal approaches have been effective in decreasing complications, assisting in earlier discharge, and in reducing hospital costs.¹⁰ Multimodal anesthesia regimens have historically consisted of short-acting opioids during general anesthesia, regional anesthesia such as epidural, and mixed-receptor medications such as ketamine and dexmedetomidine during the operation.^{10,11} The addition of intraoperative methadone to multimodal anesthesia has been shown to reduce postoperative pain, LOS, and the quantity of opioid administered in the postoperative period when compared to epidural anesthesia.¹⁰ The purpose of this study was to investigate whether adding intraoperative methadone to a multimodal pain control regimen would reduce postoperative opioid requirement in addition to decreasing LOS among patients undergoing MIRPE.

Methods

Ethical Considerations

The study protocol was reviewed and approved by the institutional review board of the University of Tennessee Health Science Center College of Medicine Chattanooga/Erlanger Health System prior to conduction.

Design and Setting

This study was an unmatched retrospective cohort study of patients who underwent MIRPE and were admitted to the pediatric ICU after surgery at a tertiary pediatric medical center in the United States. This pediatric medical center has a 15-bed pediatric intensive care unit. The center performs an average of 14 MIRPE surgeries annually.

Inclusion and Exclusion Criteria

The electronic medical record was used to review the charts of 41 patients who underwent MIRPE performed by a single pediatric surgeon. Inclusion criteria were all patients who underwent MIRPE at The Children's Hospital at Erlanger Medical Center between September 2016 and August 2020. Exclusion criteria included history of developmental delay, hepatic or renal dysfunction (bilirubin > 2.0 mg/dL or creatinine > 1.5 mg/dL), and history of substance abuse.

Analgesia Protocol

Prior to January 2020, all patients received paravertebral blocks with continuous infusion of bupivacaine which was weaned on postoperative day (POD) 1 and then discontinued on POD 2. After January 2020, patients received erector spinae plane (ESP) and intercostal blocks of liposomal bupivacaine bilaterally.

All patients were instructed to apply a scopolamine patch the night before surgery. Additionally, all were given intravenous acetaminophen for 3 doses, intravenous ketorolac every 8 hours for 3 days, hydromorphone or morphine patient-controlled analgesia (PCA), and diazepam scheduled with as-needed (PRN) doses available. Medications were transitioned to an oral regimen on POD 1 with weaning of the PCA and the addition of oxycodone-acetaminophen scheduled and PRN. Oral ibuprofen was started after discharge when ketorolac dosing was complete. Medications were adjusted based on physical exam, reported pain level, and PRN usage.

MIRPE Surgical Protocol

Operating equipment was not opened until appropriate ESP and intercostal blocks were performed. Foot traffic in and out of the operating room was prohibited after surgical equipment was opened to decrease risk of infection. Each MIRPE was performed in the standard fashion. The number of bars placed was determined intraoperatively by the surgeon. All bars were hand bent in situ by the surgeon to prevent proud hardware. Forced sternal elevators were used in the majority of cases due to the severity of defect being corrected to provide the safest plane between the chest wall and pericardium. Bars were secured with transverse fixators and fiber wire unilaterally on the lateral most aspect of the bars under videoscopic visualization. The bars were further secured using horizontal mattress polydioxanone suture in Witzel fashion.

Data Collection

Demographic data collected were as follows: age at time of surgery, weight, height, BMI, biological sex, education

level, primary language, and employment. Other data collected included surgery date, length of surgery, number of bars placed for reconstruction, type of neurologic blockade administered, Haller index, medication allergies, tobacco use, alcohol use, history of substance abuse, American Society of Anesthesiologists category, postoperative nausea or vomiting, respiratory depression during hospitalization, dose of non-methadone intraoperative opioids administered, dose of intraoperative methadone, total opioid consumption during hospital stay, hospital LOS, pain scores, time to ambulation, and time to toleration of solid food.

Analysis

Patients were stratified into 2 groups: multimodal anesthesia (MM) and multimodal anesthesia with the addition of intraoperative methadone (MM + M). The intraoperative methadone administered to patients was administered at a dose of .1 mg/kg and was given at the induction of the operation. All opioid totals (in morphine milligram equivalent [MME]) were divided by the respective body weight (kg) of the patient. Data were recorded in an Excel spreadsheet and analyzed with IBM SPSS Statistics v27. Normality was assessed by Shapiro-Wilk's test ($P > .05$) and homogeneity of variances by Levene's test ($P > .05$). Two-tailed independent-samples t -test or Mann-Whitney U test were used to determine statistically significant differences in continuous variables. Chi-square test or Fisher's exact test were used to determine statistically significant associations in categorical variables. Results are expressed with mean $\pm$ SD, median (interquartile range), 95% confidence intervals, effect size according to Cohen's d , and a P value $< .05$ is considered statistically significant.

Results

Results are summarized in Table 1. No patients met exclusion criteria. The MM group consisted of 20 patients with a median age 16 years (15-17 years), average BMI 19.88 ± 3.43 kg/m², and 70% (14/20) were male. The MM + M group consisted of 20 patients with a median age 16 years (16-17 years), average BMI 19.30 ± 1.98 kg/m², and 75% (15/20) were male. The MM group initially consisted of 21 patients; however, 1 extreme outlier was removed for having a total opioid dose during hospitalization MME/kg greater than 4.70 and postoperative MME/kg greater than 4.0, exceeding 3 SDs above the mean.

Analysis determined the addition of intraoperative methadone for patients undergoing MIRPE significantly reduced the postoperative opioid requirement (MME/kg) during admission. Postoperative MME/kg was significantly less in the MM + M group ($1.16 \pm .44$) compared to

the MM group ($1.61 \pm .55$) by .45 MME/kg (95% CI, $-.77$ to $-.13$), $t_{38} = -2.866$, $P = .007$, $d = -.906$. Additionally, hospital opioid (non-methadone) total MME/kg was significantly less in the MM + M group ($1.37 \pm .46$) compared to the MM group ($1.87 \pm .54$) by .50 MME/kg (95% CI, $-.83$ to $-.18$), $t_{38} = -3.149$, $P = .003$, $d = -.996$. There were no significant differences in average hospital LOS ($P = .068$), average pain score ($P = .091$), time to toleration of solid food ($P = .947$), nor time to ambulation ($P = .174$) between groups.

In the MM group, 25% (5/20) of patients experienced respiratory depression compared to 35% (7/20) of patients in the MM + M group. The MM group encountered nausea/vomiting in 25% (5/20) of patients and the MM + M group encountered nausea/vomiting in 35% (7/20) of patients. However, since there was no difference in time to toleration of solid food between the 2 groups, the difference in prevalence of nausea/vomiting is not considered clinically relevant (nor is it statistically significant). Patients had either 1 or 2 bars placed as determined intraoperatively by the surgeon.

Postoperative complications were primarily those that were expected and could be conservatively managed; however, 2 of the patients included in the study required more invasive interventions. One patient experienced a bar flip 3 months postoperatively as a result of non-compliance with activity restrictions as prescribed by the surgeon. Due to the occurrence of this complication 3 months postoperatively, this complication was not included in the data analysis of the study. Most MIRPE patients have some degree of pneumothorax postoperatively that resolves without tube thoracostomy placement or supplemental oxygen. One patient in the study experienced a right sided pleural effusion postoperatively which required thoracoscopy with evacuation. However, the hospital LOS for this patient was among the shortest LOS of patients (meeting inclusion criteria) at 3 days. It is possible that this patient's pleural effusion may have affected the LOS, but this would be impossible to definitively determine. No patients included in the study developed infections during their hospital admission.

Discussion

Multimodal pain control for the postoperative pain management of patients undergoing major surgery has been instituted in nearly all fields of surgery. Patients undergoing MIRPE are known to experience significant postoperative pain.^{6,10,12} Historically, our institution utilized a combination of opioids, benzodiazepines, and a variety of regional blocks to control postoperative pain. In 2018, intraoperative methadone was incorporated with the hypothesis that it would reduce postoperative narcotic requirements and reduce hospital LOS. This decision was

Table 1. Summary of Demographics and Outcomes Among Groups.^a

	Methadone (n = 20)	No methadone (n = 20)	P value ^b	Cohen's d
Age	16 (16-17)	16 (15-17)	.242	—
Gender				
Male	15 (75.0)	14 (70.0)	.723	—
Female	5 (25.0)	6 (30.0)		—
Tobacco	1 (5.0)	1 (5.0)	1.00	—
Nausea and vomiting	7 (35.0)	5 (25.0)	.490	—
Oversedation	2 (10.0)	0 (0)	.487	—
Fever	1 (5.0)	5 (25.0)	.182	—
Respiratory depression	5 (25.0)	7 (35.0)	.490	—
BMI	19.30 ± 1.98	19.88 ± 3.43	.521	—
Haller index	4.21 (3.5-5.0)	3.87 (3.3-5.0)	.336	—
ASA	2 (2.0-2.0)	2 (2.0-2.0)	.478	—
Blocks	3 (2.25-4.0)	2 (2.0-4.0)	.174	—
TOD MME	109.62 ± 25.43	115.43 ± 42.44	.602	—
TOD w/o methadone MME	79.12 ± 27.04	115.43 ± 42.44	.003	−1.021
Postoperative MME	67.12 ± 25.76	99.91 ± 41.69	.005	−.946
TOD MME/kg	1.89 ± .46	1.87 ± .54	.886	—
TOD w/o methadone MME/kg	1.37 ± .46	1.87 ± .54	.003	−.996
Postoperative MME/kg	1.16 ± .44	1.61 ± .55	.007	−.906
Length of stay (days)	3 (3.0-4.0)	4 (3.0-4.0)	.068	—
Ambulation (days)	1 (1.0-1.0)	1 (1.0-2.0)	.174	—
Tolerating solid food (days)	1 (1.0-2.0)	1 (1.0-2.0)	.947	—
Average pain score	3.16 ± 1.49	2.37 ± 1.40	.091	—

Abbreviations: ASA, Status, American, Society of Anesthesiologists Status; BMI, Body Mass Index; MME, Morphine Milligram Equivalents; TOD, Total Opioid Dose.

^aData are presented as number (%) for categorical variables, and mean ± SD, or median (interquartile range) for continuous variables.

^bHighlighted P values are statistically significant with strong effect sizes.

based on the plethora of literature that indicates that the use of intraoperative methadone safely reduces postoperative pain.¹³⁻¹⁸ The rationale for the effectiveness of intraoperative methadone is based in its pharmacokinetics.^{13,19} Methadone has a rapid onset of action (6-8 minutes) and the longest half-life (24-36 hours) of all currently used opioids.^{13,20} It is a potent N-methyl-D-aspartic acid receptor antagonist, promoting hyperalgesia and opioid tolerance.^{21,22} Additionally, methadone inhibits the reuptake of the neurotransmitters norepinephrine and serotonin, providing mood stimulation postoperatively.^{21,22} Methadone is predominantly cleared by cytochrome P450 enzymes to an inactive byproduct and a small amount of unchanged methadone is filtered and excreted in urine.^{21,22} Therefore, pharmaceuticals which affect cytochrome P450/CYP2B6 may influence the plasma concentration of methadone.^{21,22} The literature indicates adolescents metabolize methadone similarly to adults.²⁴

Methadone does not demonstrate the sharp peaks and troughs in serum concentration which are commonly associated with shorter acting opioids. This partially explains its less addictive potential compared to short-acting opioids.^{13,23} Furthermore, there is evidence to

suggest that methadone attenuates opioid tolerance and hyperalgesia which likely explains the significant difference in postoperative opioid requirements for MIRPE patients; this is meaningful in terms of reducing medically induced opioid addiction.^{16,24} This study has validated that intraoperative methadone, when combined with multimodal pain management strategies, can reduce the postoperative narcotic requirement in patients undergoing MIRPE without significant change in reported pain scores. However, a significant reduction in LOS was not observed. Advantages of a single-surgeon study include consistency in care strategies, surgical technique, decision-making, and preoperative and postoperative education.²⁵

Postoperative anesthetic options that are currently in use for pain control after MIRPE include thoracic epidurals, paravertebral blocks, intercostal blocks, elastomeric pain infusion catheters, and cryoablation.^{10-12,26,27} Effectiveness of these modalities appears to be practitioner dependent. For example, failure rates of epidural catheters are reported between 5 and 30%.¹² With the utilization of long-acting injectable anesthetics, such as liposomal bupivacaine, our practice no longer uses epidural catheters in favor of ultrasound-guided regional anesthesia of ESP and intercostal blocks.

Recent studies regarding intercostal cryoablation during MIRPE have demonstrated similar success in reducing LOS and opioid requirements with good outcomes.²⁸⁻³⁰ However, cryoablation intentionally destructs axonal nerves and carries risks such as the development of neuroma and chronic pain syndrome. In addition, cryoablation requires additional technical training not required by the simple addition of methadone to the multimodal pain regimen. Therefore, the use of cryoablation has not yet been adopted by our surgical team.

The outcomes of this study reinforce the existing literature which supports the use of intraoperative methadone in decreasing postoperative narcotic requirements. However, the reduced LOS as seen in Singhal et al study was not observed in this study.¹⁰ This could be explained by the difference in sample size between the 2 studies (n = 40 vs n = 124).

Overall, this study was a single-surgeon, single-center, unmatched retrospective cohort review of the medical records of patients meeting inclusion criteria, and the authors acknowledge the limitations and potential biases inherent to the study design. The sample size is limited by the infrequent nature of the procedure. The transition from catheter-based regional anesthesia to ultrasound-guided nerve blocks was not standardized among the patients studied. Other potential variability may exist due to differences in nursing as it relates to the recording and interpretation of pain scores. Consistency in documentation of PCA administration was variable among all included patients. The same opioid was not used in the PCA of all patients; some patients received morphine while others received hydromorphone. Finally, data collected for patients prior to August 2017 were retrieved from paper documentation rather than electronic documentation which may limit accuracy and comparative analysis.

Future research avenues involve expanding the scope of the administration of intraoperative methadone to monitor any effect on narcotic requirements after discharge in a larger cohort of surgical patients. Also, standardizing regional blocks in future studies will assist in identifying which combination of regional blocks, methadone, and multimodal pain control medications is most effective in decreasing postoperative pain and narcotic requirements.

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Declaration of Conflicting Interests

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